

ADWAITH SANTHOSH

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Work Authorization: F-1 OPT/CPT (STEM extension eligible)

SUMMARY

Machine Learning / AI Engineer with 4+ years delivering production ML and generative AI systems. Specialized in agentic AI, retrieval-augmented generation (RAG), LLM fine-tuning, and end-to-end MLOps on AWS and Azure.

EDUCATION

University of Maryland, College Park — M.S., Data Science College Park, MD • Aug 2024 – May 2026
GPA: 3.97 (Top 5%) • Coursework: Advanced Machine Learning, Deep Learning, Natural Language Processing, Big Data Systems

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C/C++

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, CNNs, RNNs/LSTMs, Graph Neural Networks (GNNs), NLP, Computer Vision, Recommender Systems

Generative AI & LLMs: LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, LLM Fine-Tuning, OpenAI, Claude, Llama, Gemini, Multimodal Models

Vector & Databases: Qdrant, Pinecone, Weaviate, OpenSearch, pgvector, PostgreSQL, MySQL, MongoDB, Redis

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, EC2, S3, RDS), Azure (ML Studio, OpenAI, AKS), GCP (Vertex AI, Cloud Run), Docker, Kubernetes, MLflow, Airflow, CI/CD, Terraform

Data & Frameworks: FastAPI, Flask, Django, PySpark, Pandas, NumPy, Kafka, Celery, Power BI, Tableau

PROFESSIONAL EXPERIENCE

AIB Inc — Machine Learning Engineer (CPT) Dallas, TX • Jan 2026 – May 2026
• Developed deep learning and SVM-based NLP models for text classification across internal products, replacing manual categorization with automated tagging and improving labeling consistency.

CertbuddyAI Inc — Machine Learning Engineer (CPT) Dallas, TX • Jun 2025 – Aug 2025
• Built an agentic multimodal conversational AI combining probabilistic and deterministic pipelines with an OpenSearch vector-DB RAG engine (S3-integrated, dynamic partitioning), enabling dynamic yet fact-checked reasoning.

University of Maryland — Dept. of Public Health — Graduate Research Assistant College Park, MD • Oct 2024 – Jan 2026
• Conducted large-scale sentiment analysis, data scraping, and cleaning across Twitter, Reddit, Bluesky, and Facebook to extract public-health insights; findings contributed to published research in *Big Data & Society*.

GoML (AWS GenAI Partner) — Machine Learning Engineer Coimbatore, India • Feb 2024 – Aug 2024
• Built a RAG chatbot for an EV-certification client using QdrantDB, AWS Bedrock (Claude 3.5 Sonnet), FastAPI, and S3 — achieving 95% retrieval accuracy, cutting document-retrieval time by 40%, and reducing API latency by 35%.

Tata Consultancy Services — Machine Learning Engineer Chennai, India • Jul 2021 – Sep 2023
• Built and deployed deep-learning session-based and hybrid (LightFM) recommender systems on Azure ML, achieving 10 ms API latency (10x faster), lifting conversion rates by 20%, and cutting model runtime 65% (7 hrs to 2.5 hrs).

PROJECTS

RAG Pipeline — Dual-Mode Document Processing System <https://rag-pipeline-zeta.vercel.app>
Python, FastAPI, QdrantDB, Groq, Llama 3.3 70B, Gemini 2.0 Flash, Supabase, React
• Built a dual-pipeline RAG system: a basic pipeline (PyMuPDF, fixed-size chunking) and an advanced multimodal pipeline (Docling) for table extraction, image analysis, and semantic chunking.
• Implemented hybrid retrieval across dual Qdrant collections with re-ranking and score boosting, evaluated on faithfulness, answer relevancy, Precision@K, MRR, and hit-rate.

RecruitSquad — Autonomous Multi-Agent Recruiting Platform <https://recruit-squad-7d8e1.web.app>
Python, FastAPI, LangGraph, OpenAI, Gemini, Firebase, React, Google Cloud Run
• Designed a 7-agent LangGraph orchestration system (sourcing, OA generation, behavioral chat, scoring, market analysis, interview coordination, audit) spanning 4 graphs with deterministic pipeline routing.
• Built a 100-point composite scoring engine integrating resume-JD fit, OA, behavioral, and multi-round interview feedback; deployed on Google Cloud Run with Firebase-hosted frontend.

RESEARCH & PUBLICATIONS

- A Decade of Discourse: Exploring Sentiments and Trends around Immigration on Social Media (2014–2024) — *Big Data & Society*.
- Developing an AI-Powered Question-and-Answer Chatbot with English–Spanish Capabilities for New Mothers — *Big Data & Society*.